Part VI — Heat, Temperature, and Thermodynamics · Chapter 34

Heat Engines, Entropy, and Efficiency

How thermal energy drives machines, why efficiency is fundamentally limited, and what entropy costs every real process

> *Every coal-fired power plant, every internal combustion engine, every jet turbine, and every nuclear reactor is, at its heart, a device that takes heat from a hot source and converts some of it into useful work. The fraction converted is the thermal efficiency — and no matter how clever the engineering, no matter how perfect the materials, the Second Law of Thermodynamics sets an absolute ceiling on that fraction that no engineer in history has breached. The world runs on thermal machines: roughly 80% of global electricity comes from heat engines of one kind or another. Understanding why those engines cannot be made perfectly efficient — and what happens to the energy that is not converted to work — is not just academic. It is the central challenge of global energy policy.*

34.1 What a Heat Engine Is

A heat engine is any device that converts thermal energy into mechanical work by exploiting a temperature difference. The engine absorbs heat from a high-temperature source (the hot reservoir), converts part of that heat into work, and rejects the remainder as waste heat to a low-temperature sink (the cold reservoir).

Heat engines are cyclic: after one complete cycle, the working fluid (gas, steam, refrigerant) returns to its original state, ready to repeat the process. Because the internal energy of the working fluid is a state function and returns to its original value after each cycle, the net work output over one cycle equals the net heat input minus the heat rejected: $W_{net} = Q_H - Q_C$.

Examples of heat engines: steam turbines (coal, gas, nuclear power plants), internal combustion engines (cars, trucks, aircraft), gas turbines (jet engines, gas-fired power plants), and Stirling engines (used in some solar power systems and submarines). Despite their enormous diversity of design, all heat engines obey the same thermodynamic constraints.

> **Key Point:** A heat engine is not a violation of conservation of energy. It takes in more heat from the hot reservoir than it delivers as work; the difference is the waste heat

rejected to the cold reservoir. Energy is conserved: $Q_H = W_{out} + Q_C$.

34.2 Energy Flow in Thermal Machines

The energy flow through a heat engine can be represented by a simple diagram. Heat Q_H flows from the hot reservoir at temperature T_H into the engine. The engine converts part of this to net work W_{out}. The remainder, $Q_C = Q_H - W_{out}$, is rejected as waste heat to the cold reservoir at temperature T_C.

Three conservation relationships define the engine:

$$Q_H = W_{out} + Q_C$$

Energy conservation for a heat engine. All input heat Q_H must appear as either work output W_{out} or waste heat Q_C. No energy is created or destroyed.

This is the First Law applied to a complete cycle ($\Delta U_{cycle} = 0$). Every joule that enters as heat from the hot reservoir is accounted for: it either exits as mechanical work or as heat rejected to the cold reservoir. The Second Law then further constrains how much of Q_H can become W_{out} — it cannot be all of it.

For a refrigerator or heat pump, the same three quantities apply, but the direction of all flows is reversed: work W_{in} is done on the system, heat Q_C is absorbed from the cold reservoir, and heat $Q_H = Q_C + W_{in}$ is rejected to the hot reservoir. The energy balance is the same; only the direction changes.

34.3 Heat Reservoirs and Thermal Cycles

A heat reservoir is an idealised object large enough that it can absorb or release any amount of heat without changing its temperature. The ocean, the atmosphere, and large geological formations act approximately as thermal reservoirs on engineering timescales.

A thermal cycle is the sequence of thermodynamic processes that the working fluid undergoes before returning to its initial state. The Carnot cycle — the theoretically optimal cycle — consists of four reversible processes: isothermal expansion at T_H (absorbing Q_H), adiabatic expansion (cooling from T_H to T_C), isothermal compression at T_C (rejecting Q_C), and adiabatic compression (warming from T_C back to T_H). Because all four processes are reversible, the Carnot cycle produces zero entropy and achieves the maximum possible efficiency.

Real engine cycles deviate from Carnot because real processes are irreversible. The

Otto cycle (petrol engines) uses adiabatic compression and expansion alternating with constant-volume heating and cooling. The Diesel cycle uses adiabatic compression and constant-pressure heating. The Rankine cycle (steam power plants) uses boiling and condensation of a working fluid. All real cycles produce entropy and have efficiency below the Carnot limit.

34.4 Work Output from Heat Engines

The net work output of a heat engine per cycle equals the heat absorbed from the hot reservoir minus the heat rejected to the cold reservoir:

$$W_{out} = Q_H - Q_C$$

Net work output of a heat engine per cycle. All three quantities are positive by convention; W_out is the useful mechanical energy delivered per cycle.

In a P-V diagram of a thermodynamic cycle, the net work output equals the area enclosed by the cycle. Clockwise cycles (on a P-V diagram) represent heat engines producing positive work. Counterclockwise cycles represent refrigerators or heat pumps consuming work.

Example — Work from heat engine A heat engine absorbs Q_H = 5000 J from a hot reservoir and rejects Q_C = 3500 J to a cold reservoir per cycle. Net work output: W_out = 5000 – 3500 = 1500 J. Efficiency: η = 1500/5000 = 30%.

34.5 Thermal Efficiency

Thermal efficiency η (eta) is the fraction of heat input that is converted to useful work output. It is the primary figure of merit for a heat engine:

$$\eta = W_{out} / Q_H = 1 - Q_C / Q_H$$

Thermal efficiency. W_out = net work output; Q_H = heat absorbed from hot reservoir. η ranges from 0 (no useful work) to 1 (100% conversion, forbidden by Second Law).

Efficiency is always less than 1 (less than 100%) for any real engine operating between finite temperatures. The Second Law prohibits η = 1 (Q_C = 0), because that would require all heat from a single reservoir to be converted to work — the Kelvin-Planck statement of the Second Law.

Real engine efficiencies: a modern coal power plant achieves about 40–45%. A combined-cycle gas turbine (CCGT) can reach 60–62% by cascading two thermodynamic cycles. A modern petrol car engine is about 25–35% efficient. A diesel engine is about 35–45%. Human skeletal muscle is about 25% efficient as a mechanical engine. The rest of the input energy becomes waste heat in every case.

Engine Type	Typical T_H	Typical T_C	Carnot Limit	Actual Efficiency
Coal steam turbine	500°C (773 K)	30°C (303 K)	61%	38–42%
CCGT gas turbine	1500°C (1773 K)	30°C (303 K)	83%	58–62%
Nuclear steam turbine	300°C (573 K)	30°C (303 K)	47%	30–35%
Petrol car engine	2000°C (2273 K)*	20°C (293 K)	87%*	25–35%
Diesel engine	2000°C (2273 K)*	20°C (293 K)	87%*	35–45%

Table 34.1 *Efficiency comparison for common heat engines. *Combustion temperature is peak value; effective T_H is lower due to cycle irreversibilities.*

34.6 Why No Engine Can Be 100% Efficient

The impossibility of 100% thermal efficiency follows directly from the Second Law. An engine operating at 100% efficiency would convert all Q_H into work and reject nothing (Q_C = 0). For a cycle, this requires the engine to absorb heat from a single reservoir and convert it entirely to work — exactly the process the Kelvin-Planck statement prohibits.

The Carnot analysis provides the fundamental limit. For the Carnot cycle, entropy changes of the hot and cold reservoirs must be equal and opposite (no net entropy change, since the cycle is reversible):

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ΔS_H = Q_H/T_H   (entropy removed from hot reservoir)
ΔS_C = Q_C/T_C   (entropy added to cold reservoir)
For Carnot: ΔS_H = ΔS_C  ⇒  Q_H/T_H = Q_C/T_C  ⇒
Q_C/Q_H = T_C/T_H
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Carnot entropy balance. For a reversible cycle, entropy produced equals zero: entropy removed *from hot reservoir equals entropy added to cold reservoir.*

Substituting into the efficiency formula: $\eta_{Carnot} = 1 - Q_C/Q_H = 1 - T_C/T_H$. This is the maximum efficiency between any two temperatures. To achieve $\eta = 1$, you need $T_C = 0$ K, which the Third Law forbids.

For any real (irreversible) engine: $Q_C/Q_H > T_C/T_H$, meaning Q_C is larger than the Carnot minimum. More energy is wasted as heat than necessary, and efficiency is below Carnot. Irreversibility always costs efficiency.

> **Example — Carnot vs. actual efficiency** A steam plant operates between T_H = 600°C (873 K) and T_C = 30°C (303 K). Carnot limit: $\eta_C = 1 - 303/873 = 65.3\%$. Actual efficiency: 42%. The actual engine wastes 23.3% more of Q_H than the theoretical minimum.

34.7 Refrigerators and Heat Pumps

A refrigerator is a heat engine operating in reverse. Work W_{in} is supplied externally; heat Q_C is extracted from the cold reservoir (inside the fridge); heat $Q_H = Q_C + W_{in}$ is rejected to the warm reservoir (the room). The working fluid (refrigerant) is compressed, condenses in the exterior coils (releasing Q_H), expands through a valve, and evaporates in the interior coils (absorbing Q_C). The Second Law requires that $W_{in} > 0$: you cannot move heat from cold to hot spontaneously.

A heat pump is a refrigerator whose purpose is to deliver heat Q_H to the warm side (to heat a building), rather than to extract heat Q_C from the cold side. Both devices obey the same energy balance: $Q_H = Q_C + W_{in}$. The distinction is only in which direction the heat flow is useful.

The energy efficiency of refrigerators and heat pumps is measured by the coefficient of performance (COP) rather than thermal efficiency, because the “useful output” is heat transfer, not work.

34.8 Coefficient of Performance (Conceptual)

The coefficient of performance (COP) measures how much useful heat transfer you get per unit of work input. For a refrigerator, the useful output is Q_C (heat removed from the cold space); for a heat pump, it is Q_H (heat delivered to the warm space):

$$COP_{refrigerator} = Q_C / W_{in} = Q_C / (Q_H - Q_C)$$

COP for refrigerator. How much heat is extracted from the cold side per joule of work input. Carnot maximum: $COP_C = T_C/(T_H - T_C)$.

$$COP_{heat\ pump} = Q_H / W_{in} = Q_H / (Q_H - Q_C)$$

COP for heat pump. How much heat is delivered to the warm side per joule of work input. Carnot maximum: $COP_{HP} = T_H/(T_H - T_C)$. Note: $COP_{HP} = COP_{refrigerator} + 1$.

COP values greater than 1 are normal and expected — a heat pump can deliver 3–5 J of heat per joule of electrical work because it is moving heat, not generating it. This is why heat pumps are far more energy-efficient for space heating than electric resistance heaters (which achieve exactly COP = 1). A ground-source heat pump exploiting the stable ground temperature (about 10–12°C) can have a COP of 4–5 in winter heating conditions.

Example — COP example A refrigerator extracts $Q_C = 350$ J per cycle and requires $W_{in} = 150$ J of work. $Q_H = 350 + 150 = 500$ J rejected to the room. $COP_{refrigerator} = 350/150 = 2.33$. Carnot COP ($T_C = 5°C = 278$ K, $T_H = 30°C = 303$ K): $COP_C = 278/(303 - 278) = 278/25 = 11.1$. The real refrigerator has COP well below the Carnot limit.

34.9 Entropy Revisited

Chapter 33 introduced entropy as a measure of microscopic disorder and as the thermodynamic state function that quantifies the Second Law. In the context of heat engines, entropy has a very concrete operational meaning: it measures the quality of energy, and specifically how much of the thermal energy in a reservoir is available to do work.

High-temperature reservoirs have low entropy per joule (Q/T_H is small for large T_H) — their energy is "high quality" and more of it can be converted to work. Low-temperature reservoirs have high entropy per joule (Q/T_C is large for small T_C) — their energy is "low quality" and little can be converted to work.

When a heat engine operates, it transfers entropy from the hot reservoir (where energy has low entropy) to the cold reservoir (where energy has high entropy). The engine must reject at least enough heat to the cold reservoir to carry away the entropy it extracted from the hot reservoir. This is the entropy cost of doing work: you cannot do work without also increasing the entropy of the universe.

The work that an engine extracts is the energy that does not appear as entropy transfer. This is the exergy (or available work) concept: only the fraction of Q_H that exceeds the minimum entropy transfer requirement can be converted to work. This is exactly the Carnot fraction $1 - T_C/T_H$.

34.10 Entropy and Irreversibility

Every real process is irreversible to some degree: friction generates heat, heat flows across finite temperature differences, and mixing occurs spontaneously. Each of these irreversibilities produces entropy, and that extra entropy must be rejected to the environment as additional waste heat. This additional rejection reduces the work that can be extracted from a given heat input.

Sources of irreversibility in heat engines: (1) heat transfer across finite temperature differences at the hot and cold ends; (2) friction in moving parts; (3) throttling of fluids through valves; (4) combustion and chemical reactions (intrinsically irreversible); (5) turbulence in fluid flow. Each source produces entropy and reduces efficiency below the Carnot limit.

The degree of irreversibility of a process is quantified by its entropy generation ΔS_{gen}. For a reversible process, $\Delta S_{gen} = 0$. For any real process, $\Delta S_{gen} > 0$. The efficiency penalty due to irreversibility is $\eta_{Carnot} - \eta_{actual} = (T_C/Q_H)\ \Delta S_{gen}$. Reducing irreversibility — by decreasing temperature differences, improving insulation, minimising friction — always improves efficiency.

Modern thermodynamic analysis of power plants uses “exergy accounting” or “second-law analysis” to identify which components are most irreversible and therefore where engineering improvements would have the greatest effect. A conventional exergy analysis of a coal power plant typically shows that the boiler (where hot combustion gases transfer heat to steam across a large temperature difference) is responsible for 50–60% of all entropy production — making it the primary engineering target.

34.11 Waste Heat and Real-World Engineering Limits

Waste heat is the heat rejected to the cold reservoir (Q_C). In all heat engines, it is unavoidable — it is the price paid for extracting work. The global scale of waste heat production is staggering: global electricity generation is about 25 000 TWh/yr at 35% average efficiency, implying about 46 000 TWh/yr of waste heat is rejected to rivers, oceans, and the atmosphere.

Power plant cooling is a major engineering and environmental challenge. Large thermal power plants reject 60–70% of their fuel energy as waste heat. Most use cooling towers (evaporative cooling with atmosphere) or once-through cooling with river or ocean water. The thermal pollution of rivers by warm water discharge can damage aquatic ecosystems; regulations in many countries limit the temperature rise permitted.

Combined heat and power (CHP or cogeneration) addresses waste heat by using the rejected Q_C for space or industrial process heating rather than discarding it. A CHP plant operating at 35% electrical efficiency that also uses 45% of fuel energy as useful heat achieves an overall useful energy fraction of 80% — more than double the electrical-only efficiency.

Transportation heat engines (cars, planes, ships) cannot easily use their waste heat, so it is simply discharged. This is one of the fundamental thermodynamic inefficiencies of the transport sector: a petrol car delivers about 20–25% of fuel energy to the wheels; the rest heats the engine, exhaust, brakes, and tyres. Electric vehicles do not use heat engines for propulsion; they convert electrical energy to mechanical energy with about 90% efficiency, subject only to frictional and electrical losses, not the Carnot constraint.

34.12 Energy Systems and Thermodynamic Optimisation

The global energy challenge is fundamentally a thermodynamic challenge: how to extract maximum useful work from finite energy resources while minimising entropy production and waste. Thermodynamic optimisation encompasses several strategies.

Raising T_H

The Carnot efficiency increases as T_H increases. This is why advanced power plants use ultra-supercritical steam at 600–700°C and 30+ MPa — much hotter and higher-pressure than conventional plants. Materials capable of withstanding these conditions (nickel superalloys, ceramic coatings) are among the most technically demanding in industrial use.

Lowering T_C

Reducing the cold reservoir temperature also improves efficiency. In practice, T_C is limited by the available cooling medium (river, ocean, or atmosphere) and the cost of refrigerating below ambient. Combined-cycle plants cascade a high-temperature gas turbine (T_C of first cycle $\approx$ 500°C) with a lower-temperature steam turbine ($T_C \approx$ 30°C), effectively extending the temperature range and dramatically improving overall efficiency.

Reducing irreversibility

Even between fixed T_H and T_C, real engines fall short of Carnot efficiency because of irreversible processes. Improving turbine and compressor aerodynamics (reducing flow losses), reducing heat transfer temperature differences (larger heat exchangers), and minimising mechanical friction all reduce entropy production and improve efficiency. Modern gas turbine blades use sophisticated internal cooling channels to operate at gas temperatures above their melting point.

Storage and grid-scale applications

Intermittent renewable energy (wind, solar) is not always available when demand is high. Thermodynamic storage systems — pumped-hydro, compressed-air energy storage (CAES), molten-salt thermal storage at concentrated solar plants — use thermodynamic principles to store and retrieve energy. Each storage cycle involves thermodynamic losses, and second-law analysis guides the design of systems that minimise those losses.

> *"The development of efficient, clean heat engines is the central engineering challenge of the 21st century. The laws of thermodynamics tell us what is possible; engineering tells us how close we can get."* — Adapted from engineering thermodynamics literature

34.13 Common Mistakes in Efficiency Analysis

- Confusing $\eta = W\_out/Q\_H$ with W_out/Q_C. Efficiency is always work output divided by heat input from the hot reservoir, not by heat rejected. Using Q_C gives a ratio greater than 1, which is meaningless for an engine.
- Forgetting that $Q\_C = Q\_H - W\_out$ (not $Q\_C = 0$ unless the engine is perfect). Always check the energy balance: Q_H must equal $W\_out + Q\_C$. If you compute $Q\_C = 0$ for a real engine, something is wrong.
- Using Celsius instead of Kelvin in the Carnot efficiency formula. $\eta\_Carnot = 1 - T\_C/T\_H$ requires absolute temperatures. Using 30°C/600°C = 0.05 gives η = 95%, which is wildly wrong. The correct calculation is 303/873 = 0.347, giving η = 65.3%.
- Thinking a COP > 1 means energy is being created. A refrigerator or heat pump with COP > 1 is not violating conservation of energy. It is moving heat using work; the heat moved (Q_C or Q_H) can exceed the work input because thermal energy from the cold or hot reservoir contributes. No energy is created.
- Believing a more efficient engine wastes less fuel absolutely. A more efficient engine converts a higher fraction of fuel energy to work. But if the engine does more work (for example, a larger engine), its absolute fuel consumption and waste heat may be higher even if its percentage efficiency is better.
- Confusing the Carnot efficiency with the actual efficiency. Carnot is the maximum possible efficiency between given temperatures. Actual engines are always less efficient. Never use η_Carnot as the expected efficiency of a real engine.

Common Mistake: In all Carnot and efficiency calculations, convert temperatures to Kelvin. T_C/T_H in Celsius is meaningless. The Carnot formula $\eta = 1 - T\_C/T\_H$

requires both temperatures in kelvin.

Chapter Summary

Heat engines convert thermal energy into work by operating between a hot reservoir (T_H) and a cold reservoir (T_C). The Second Law imposes an absolute upper limit on how much of the input heat can become work.

- Energy balance: $Q_H = W_{out} + Q_C$. First Law applied to a complete cycle.
- Thermal efficiency: $\eta = W_{out}/Q_H = 1 - Q_C/Q_H$. Always < 1 for any real engine.
- Carnot (maximum) efficiency: $\eta_{Carnot} = 1 - T_C/T_H$. T in kelvins. Any irreversibility reduces η below this limit.
- Refrigerators and heat pumps operate in reverse: W_{in} drives heat from cold to hot. $COP_{refrigerator} = Q_C/W_{in}$; $COP_{heat\ pump} = Q_H/W_{in}$. Both can have $COP > 1$.
- Entropy production by irreversible processes reduces engine efficiency below the Carnot limit. Friction, heat transfer across ΔT, mixing, and combustion all generate entropy.
- Waste heat (Q_C) is unavoidable; cogeneration recaptures it for useful heating, greatly improving overall system efficiency.
- Raising T_H, lowering T_C, and reducing irreversibilities are the three engineering levers for improving thermodynamic efficiency.

Key Terms

heat engine: A device that converts thermal energy into mechanical work by operating cyclically between a hot reservoir and a cold reservoir; $W_{out} = Q_H - Q_C$.

hot reservoir (T_H): The high-temperature source from which the heat engine absorbs heat Q_H; assumed large enough that its temperature is constant.

cold reservoir (T_C): The low-temperature sink to which the heat engine rejects waste heat Q_C; assumed large enough that its temperature is constant.

thermal efficiency (η): $\eta = W_{out}/Q_H = 1 - Q_C/Q_H$; the fraction of input heat converted to useful work output; always < 1 for any real engine.

Carnot efficiency (η_{Carnot}): $\eta_{Carnot} = 1 - T_C/T_H$; the maximum possible efficiency for any heat engine operating between T_H and T_C; achieved only by a fully reversible (Carnot) cycle.

Carnot cycle: The theoretically optimal reversible thermodynamic cycle consisting of two isothermal and two adiabatic processes; produces no entropy and achieves the Carnot efficiency.

refrigerator: A heat engine operating in reverse: work W_{in} moves heat Q_C from a cold reservoir to a hot reservoir; $Q_H = Q_C + W_{in}$.

heat pump: A refrigerator whose useful output is the heat Q_H delivered to the warm space; used for space heating; $COP_{HP} = Q_H/W_{in}$.

coefficient of performance (COP): The ratio of useful heat transfer to work input for a refrigerator or heat pump; $COP_{ref} = Q_C/W_{in}$; $COP_{HP} = Q_H/W_{in}$; both can exceed 1.

waste heat (Q_C): The thermal energy necessarily rejected to the cold reservoir in every heat engine cycle; unavoidable consequence of the Second Law.

exergy: The maximum useful work obtainable from a given amount of energy in a given environment; accounts for both quantity and quality (temperature) of the energy.

entropy generation (ΔS_{gen}): Entropy produced by irreversibilities in a real process; always ≥ 0; quantifies the degree of irreversibility and the resulting efficiency loss.

cogeneration (CHP): Combined heat and power: a power system that uses both W_{out} and Q_C for useful purposes, achieving overall energy utilisation of 70–85%.

Otto cycle: The idealised thermodynamic cycle of a petrol internal combustion engine; uses isochoric (constant-volume) heat addition and rejection.

Rankine cycle: The thermodynamic cycle used in steam power plants; involves boiling, steam expansion through a turbine, condensation, and pumping.

Concept Review Questions

Answer in complete sentences. No calculations required unless stated.

1. Explain, using the First Law, why no heat engine can have efficiency greater than 1 (100%). Then explain, using the Second Law, why no real engine can even achieve 100% efficiency.
2. A heat engine operates between 400 K and 300 K. Its Carnot efficiency is 25%. Explain what this means: where does the other 75% of Q_H go?

3. Explain how a heat pump can deliver 4 joules of heat per joule of electrical work without violating energy conservation.
4. Why does raising the hot-reservoir temperature T_H increase a heat engine's efficiency? Use the Carnot formula and an entropy argument.
5. Explain what entropy generation means in a real heat engine. Give three specific sources of entropy generation in a petrol engine.
6. Compare a 40% efficient power plant operating as electrical-only to a cogeneration system using the same fuel with 35% electrical efficiency and 45% thermal efficiency. Calculate the useful energy fraction in each case.
7. An electric vehicle and an equivalent petrol vehicle both travel 100 km. The petrol engine is 25% efficient; the electric motor is 90% efficient; the electricity is generated from a coal plant at 38% efficiency. Compare the total fuel energy consumed per 100 km in each case.
8. Explain why the Carnot efficiency equation requires absolute (Kelvin) temperatures. Demonstrate numerically what goes wrong if Celsius is used for $T_H = 327°C$ and $T_C = 27°C$.

Heat Engine Problems

9. A heat engine absorbs $Q_H = 8000$ J and rejects $Q_C = 5500$ J per cycle. Find (a) the net work output, (b) the thermal efficiency.
10. A heat engine operates at 30% efficiency and produces 600 J of work per cycle. Find Q_H and Q_C.
11. A Carnot engine operates between $T_H = 800$ K and $T_C = 320$ K. Find the Carnot efficiency and the heat rejected Q_C if $Q_H = 5000$ J.
12. A real steam turbine operates between $T_H = 550°C$ and $T_C = 35°C$ and achieves 38% efficiency. (a) Find the Carnot efficiency. (b) Find the second-law efficiency η/η_{Carnot} (what fraction of the maximum possible efficiency is actually achieved).
13. A heat engine produces 100 MW of electrical output at 33% efficiency. (a) Find Q_H (total heat input per second). (b) Find Q_C (heat rejected per second). (c) If the cooling water rises by 8°C, what mass flow rate of cooling water is required? ($c_{water} = 4186$ J/kg·K)

Efficiency Calculations

14. Carnot efficiency between T_H = 600 K and each of: (a) T_C = 300 K, (b) T_C = 200 K, (c) T_C = 100 K. Plot the trend and explain.
15. Two power plants use the same cold reservoir (T_C = 300 K). Plant A has T_H = 500 K; Plant B has T_H = 1000 K. (a) Find Carnot efficiency for each. (b) Doubling T_H more than doubles the efficiency. Explain why.
16. A refrigerator maintains an interior at 4°C (277 K) in a room at 25°C (298 K). The Carnot COP is $T_C/(T_H - T_C) = 277/21 = 13.2$. The actual COP is 3.5. How much work is required per cycle to remove Q_C = 200 J of heat?
17. A heat pump heats a building at 22°C (295 K) using ground-source heat at 10°C (283 K). Carnot $COP_{HP} = T_H/(T_H - T_C) = 295/12 = 24.6$. Actual COP_{HP} = 4.0. If the building needs 15 kW of heating, what electrical power is required? Compare to a 15 kW electric resistance heater.

Entropy Concept Exercises

18. A Carnot engine absorbs 2000 J from a reservoir at 600 K. Find: (a) entropy removed from hot reservoir; (b) Carnot efficiency; (c) W_{out}; (d) Q_C; (e) entropy added to cold reservoir at 300 K; (f) net entropy change of the universe.
19. An irreversible engine absorbs 2000 J from T_H = 600 K and rejects 1500 J to T_C = 300 K. Find: (a) W_{out}; (b) actual efficiency; (c) Carnot efficiency; (d) entropy added to cold reservoir; (e) net entropy change of universe. Compare to Problem 1.
20. Explain using entropy why combined-cycle power plants are more efficient than single-cycle plants. Which law of thermodynamics does the second cycle exploit?

Applied Thermodynamic Systems Problems

21. A coal power plant burns coal with energy density 29 MJ/kg and operates at 38% efficiency. (a) How much coal (kg) is needed to produce 1 GWh of electricity? (b) How much heat (GJ) is rejected to the cooling water? (c) If the plant operates 8000 hours/year, what is the annual coal consumption in tonnes?

22. A cogeneration plant produces 30 MW of electricity (at 32% efficiency from fuel) and supplies 20 MW of district heating from its waste heat. (a) What is the total fuel input power? (b) What fraction of fuel energy is converted to useful output (electrical + thermal)? (c) How does this compare to the electrical efficiency alone?
23. A refrigeration system maintains a cold store at −18°C (255 K) in a warehouse where ambient temperature is 35°C (308 K). Carnot COP = 255/53 = 4.81. Actual COP = 2.2. If the cold store leaks 5 kW of heat, what compressor power is required to maintain temperature?
24. An electric vehicle (EV) uses a motor with 92% efficiency and a battery with 97% round-trip efficiency. A petrol vehicle with the same road load uses a 28% efficient engine. Both drive 100 km requiring 15 kWh of energy at the wheels. (a) What energy does each vehicle consume from its primary source (battery for EV, fuel for petrol car)? (b) If the EV's electricity comes from a grid with average efficiency 40% from primary fuel, which vehicle uses less primary fuel energy per 100 km?

Chapter 34 Checkpoint

Before moving on to Part VII, confirm you can do each of the following.

I can...	See Section
Define a heat engine and state the energy balance $Q_H = W_{out} + Q_C$	34.1–34.2
Calculate thermal efficiency $\eta = W_{out}/Q_H = 1 - Q_C/Q_H$	34.5
Calculate Carnot efficiency $\eta_{Carnot} = 1 - T_C/T_H$ (in kelvins)	34.6
Explain why no real engine can achieve the Carnot efficiency	34.6
Describe how a refrigerator and heat pump differ from a heat engine	34.7
Calculate COP for refrigerators and heat pumps; explain why COP > 1 is possible	34.8
Explain the entropy cost of doing work; state why irreversibility wastes energy	34.9–34.10

Describe three engineering strategies for improving thermodynamic efficiency	34.12
Identify and correct common errors in efficiency and COP calculations	34.13

End of Part VI: Heat, Temperature, and Thermodynamics — Continue to Part VII: Electricity and Magnetism →